Python Programming Basics

Python is a popular programming language used for data science, web development, and automation.

It was created by Guido van Rossum and released in 1991. Visit https://www.python.org for more info.

Many students learn python because it is easy to read and has simple syntax rules.

There are 2 major versions currently in use today across many companies worldwide.

Data Structures Overview

Lists, tuples, dictionaries, and sets are the core built in data structures in python.

A list can store 100 items or more without any real limit on its size at all.

Dictionaries store key value pairs and are extremely useful for fast lookups in code.

For more details check https://docs.python.org/3/tutorial/datastructures.html today.

Working with Files

Reading and writing files is a common task in python programs across many domains.

The open() function is used with modes like read, write, and append for files.

It is good practice to close files after use or use the with statement instead of manual closing.

In 2020 many tutorials started recommending the with statement as the default approach.

Introduction to Libraries

Python has thousands of libraries available on PyPI for almost every single task imaginable.

Popular libraries include numpy, pandas, requests, and matplotlib for various different purposes.

You can install any library using pip install followed by the library name in your terminal.

Visit https://pypi.org to search for over 400000 packages available for download right now.

Python for Data Analysis

Python is widely used for data analysis because of powerful libraries like pandas and numpy. Analysts use python to clean data, build models, and create charts for business reports. Learning python well can take around 6 months of consistent practice and daily coding. Many companies now require python skills for entry level data analyst job positions today.

Conclusion and Next Steps

This document covered basic python concepts including data structures and file handling. Students should continue practicing python by building small projects on their own time. Check https://realpython.com for 50 additional tutorials covering more advanced python topics. Good luck learning python and keep practicing every single day for the best possible results.